

AIML PROJECT SYNOPSIS

CLINICAL RISK ASSESMENT FOR TYPE-2 DIABETES

Problem Id: 7

SECTION-Q

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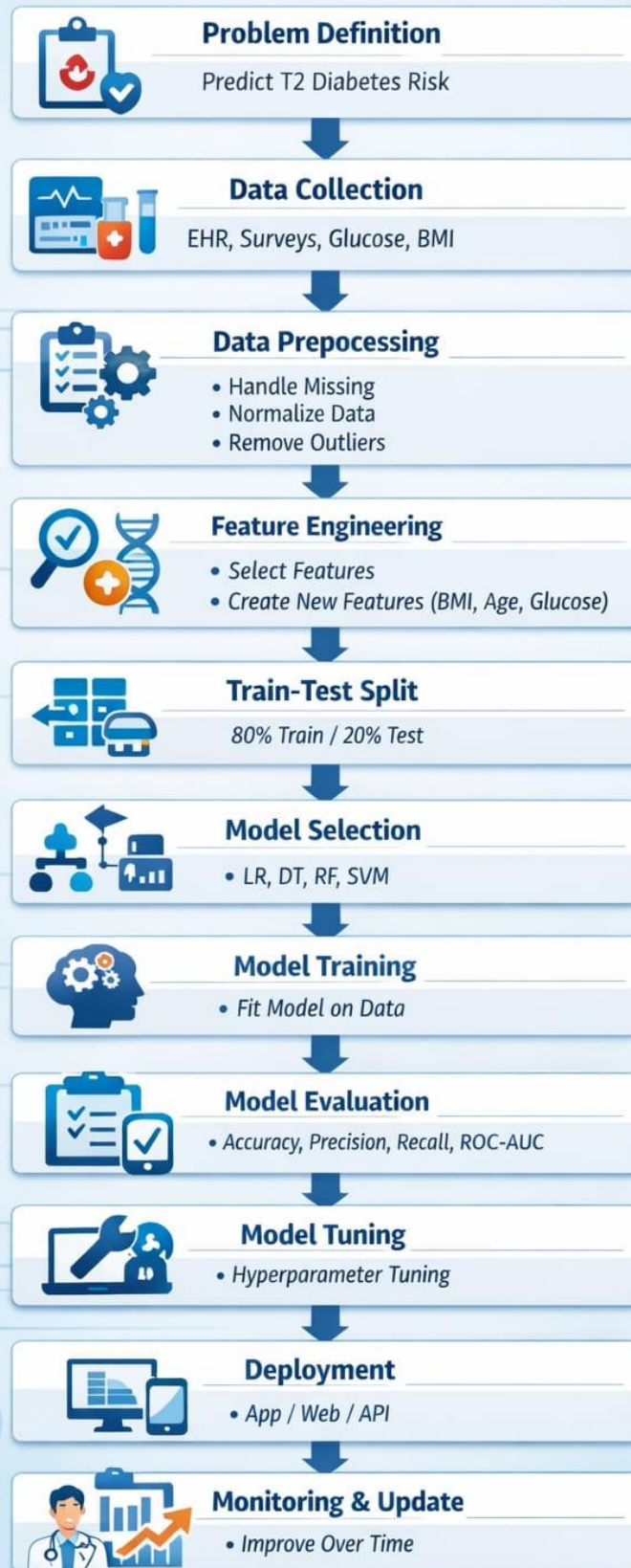
• Problem Description:

- Type-2 Diabetes is a chronic metabolic disorder that often develops silently, with long-term complications such as cardiovascular disease, kidney failure, and neuropathy. Early diagnosis is critical to reducing these risks and improving patient outcomes. Clinical risk assessment relies on measurable health indicators like Body Mass Index (BMI), insulin levels, and age, which are strong predictors of diabetes onset. In this project, students will synthesize a dataset of 250 patient records based on realistic medical ranges for these variables. The dataset will include categorical outcomes representing diabetes risk levels, which will be encoded using Label Encoding for machine learning compatibility. To ensure fair feature contribution, continuous variables will be standardized using **StandardScaler**. A **Logistic Regression classifier** will then be applied to predict risk categories. Model evaluation will emphasize the **confusion matrix**, with a particular focus on **recall**, since minimizing false negatives is crucial in healthcare—missing a high-risk patient could lead to severe complications. This pipeline demonstrates how data

synthesis, preprocessing, and classification can be integrated to support clinical decision-making and highlights the importance of recall in medical risk prediction.

- **Steps of Implementation:**

Clinical Risk Assessment for Type 2 Diabetes – AI/ML Pipeline



- **Expected Modules/ Libraries to be used :**

- **Scikit-learn (sklearn)** – Provides LabelEncoder, StandardScaler, LogisticRegression, and evaluation metrics.
- **Matplotlib** – For plotting confusion matrices and other visualizations.
- **Seaborn** – For enhanced statistical plots (e.g., heatmaps of confusion matrix).
- **Custom Module (data_synthesis.py)** – Script to generate synthetic health records with realistic ranges.
- **Custom Module (evaluation.py)** – Script to compute recall and plot evaluation metrics.
- **NumPy** – For numerical operations and random data generation.
- **Pandas** – For tabular data handling and preprocessing.

- **Expected Evaluation metrics and Plots to be used:**

- **Recall Score** – Primary metric since identifying high-risk patients is critical (minimizing false negatives).
- **Accuracy & Precision (secondary)** – For overall performance context.
- **ROC Curve & AUC** – To assess classifier discrimination ability.
- **Heatmap Plot** – For confusion matrix visualization, making results presentation-ready.
- **Confusion Matrix** – To visualize classification performance across true positives, false positives, true negatives, and false negatives

- **Required topics from the subject:**

- **Strings** – Used for handling categorical labels (“High Risk”, “Low Risk”) before encoding.
- **Pointers** – Conceptual understanding helps in memory referencing; in Python, object references act like pointers.
- **Files** – Reading/writing CSV files for storing synthetic health records and saving evaluation results.

- **Platform Required:**

- **Jupyter Notebook** – For interactive coding, visualization, and documentation.
- **Visual Studio Code** – For modular script development and debugging.
- **Python IDLE** – For quick testing of code snippets.

- **Books and Link Sources:**

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron.

2. Python Machine Learning by Sebastian Raschka and Vahid Mirjalili.

3. Data Science from Scratch by Joel Grus.

4. Official Scikit-learn Documentation-[scikit-learn: machine learning in Python — scikit-learn 1.8.0 documentation](https://scikit-learn.org/stable/index.html)